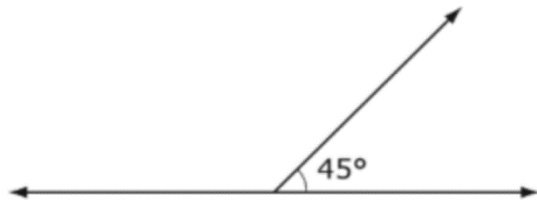
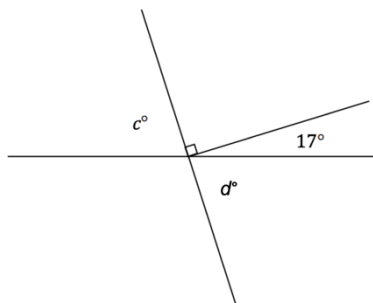


1. Two angles are supplementary. Find the missing angle measure.



$$180^\circ - 45^\circ = 135^\circ$$

Use the diagram for questions 2–3.



2. The measure of d is

- ☐ 17°
☒ 73°
☐ 163°

because they are

- ☒ complementary angles.
☐ supplementary angles.
☐ vertical angles.

3. The measure of c is

- ☐ 17°
☒ 73°
☐ 163°

because c and d are

- ☐ complementary angles.
☐ supplementary angles.
☒ vertical angles.

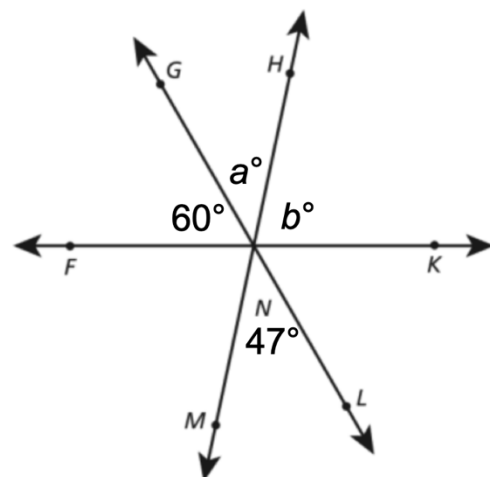
4. Three lines intersect at N . Find the value of a and b .

$$a = 47^\circ$$

$$60 + 47 + b = 180$$

$$107 + b = 180$$

$$b = 73^\circ$$



Use the following diagram for questions 5–7.

5. Write an equation that can be used to find the value of x .

$$2x - 6 = -50 + 6x$$

6. Solve the equation.

$$2x - 6 = -50 + 6x$$

$$-6 = -50 + 4x$$

$$44 = 4x$$

$$11 = x$$

7. Use the solution to find the measure of each angle.

$$\angle ACB = \underline{2(11) - 6 = 16^\circ}$$

$$\angle DCE = \underline{-50 + 6(11) = 16^\circ}$$

Use the following diagram for questions 8-10.

8. Darcy wrote the equation $x - 9 = 3x + 17$ to find the value of x , but made a mistake. What error did she make?

Darcy set the expressions equal to each other. The correct way to solve the problem is to set their sum equal to 180° .

9. Write and solve an equation that can be used to find the value of x .

$$(x - 9) + (3x + 17) = 180$$

$$4x + 8 = 180$$

$$4x = 172$$

$$x = 43$$

10. Use the solution to find the measure of each angle.

$$\angle MON = \underline{3(43) + 17 = 146^\circ}$$

$$\angle MOP = \underline{43 - 9 = 34^\circ}$$

$$\angle POQ = \underline{146^\circ}$$

